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Editorial: *Hey Google Scholar, Let's Calculate the Sharability Index!*

In the era of open science, traditional impact metrics like the h-index fail to capture one of the most critical contributions researchers make—the sharing of high-quality datasets. To address this gap, the s-index was proposed by Olfson et al. [doi: 10.1001/jamapsychiatry.2016.2610].

At the IEEE Data Reporting Best Practices Task Force, we have a vision for promoting rigorous data curation, verification, and sharing practices within the IEEE community and beyond. Our work will not only set high standards for data reporting but also highlight the immense potential of well-curated datasets to enhance reproducibility and accelerate discovery. As a testament to this vision, I have tried to be a leader in this movement by actively releasing impactful publicly accessible datasets resulting from my research.

Including the s-index on Google Scholar would signal a major step forward in recognizing the full spectrum of scholarly contributions. It would offer tangible credit for data sharability—a quality essential not only for enhancing reproducibility but also for fostering a more transparent, collaborative research culture. By spotlighting the value of well curated and widely used datasets, the s-index would incentivize researchers to invest in robust data publishing and sharing practices, thereby accelerating scientific progress.

METHOD

Mathematically, the s-index is defined as

$$s = \max\{i \in \mathbb{N} : f(i) \geq i\}$$

where $f(i)$ represents the number of citations for the ith most cited dataset. This simple formulation mirrors the familiar h-index but focuses on publicly accessible dataset sharability. It quantifies the influence of shared data by ranking datasets—usually the accompanying conference paper or journal article—in descending order of citation counts and identifying the largest number N for which at least N datasets have garnered N or more citations.

EXAMPLE

A quick glance at my Google Scholar profile shows robust citation counts for my key shared datasets summarized in Table I. Applying the s-index definition to these six datasets when arranged in descending order, we see that five of the six datasets all have at least five citations. Thus, my s-index given the data in Table I is as follows:

$$s\text{-index} = 5.$$

TABLE I. **Stats for Six Datasets as on 13 March 2025**

Dataset	DOI	Year	Citations
AMPds v1	10.1109/EPEC.2013.6802949	2013	465
AMPds v2	10.1038/sdata.2016.37	2016	368
RAE	10.3390/data3010008	2018	105
HUE	10.1016/j.dib.2019.103744	2018	69
EV-CPW	10.1109/ACCESS.2023.3340131	2023	13
SFU-EVP	10.1109/IEEEDATA.2024.3471469	2024	0

This simple calculation underlines the profound impact that data sharing can have, even when the number of datasets is modest. It also makes a persuasive case for integrating such a metric into Google Scholar profiles.

CLOSING

In summary, the s-index provides a clear, quantitative measure of data impact, reinforcing the notion that datasets are not mere byproducts but vital assets in the research ecosystem. Google Scholar's adoption of the s-index would not only celebrate and reward data sharing but also advance the principles of open science that underpin reproducibility and innovation.

It is time to recognize that in today's digital age, dataset sharability is as crucial as publication counts. *Let us promote and celebrate researchers by giving their shared data the credit it deserves.*

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